

TABLE I
Power Consumption by Home Appliances

Electrical appliance	Power (watts)
Bulb	58.5W
Tubelight	39.6W
Television (portable)	59.9W
Refrigerator	1000W initially 117W while running
Fan (ceiling)	87.9W
Computer with LCD screen & small speakers (personal computer)	79.9W with speakers, 76.9W without speakers; varies from 80W to 100W

TABLE II
Supply to LCD Computer with 12V, 120Ah Battery, Solar Inverter with UPS

Time	Output voltage (Vo)	Output current (Io)	Battery voltage
Initially	230V	15A	12.5V
1 minute	230V	17A	12.5V
2 minutes	Low voltage	Low current	10V

Fresnel reflectors design. Fresnel reflector is a new concept similar to parabolic trough design. It uses a set of flat mirrors, which have a common receiver (unlike parabolic design where each mirror has its own receiver). The advantages of fresnel reflectors are the use of flat mirrors, fewer number of receivers and lower energy station area decreasing the cost dramatically. Although the efficiency of fresnel reflectors is lower than parabolic-trough design, low cost per energy unit can make fresnel reflectors quite popular in the future. Currently, these are in developing stage.

Power consumption of home appliances

Before calculating solar power system needs of your house, you need to calculate the electricity consumption of your house. Power consumption of various home appliances is shown in Table I. These equipment work on 230V, 50Hz input voltage.

Let's assume you want power backup for electrical load of one tubelight, one fan, one TV, one computer and one bulb. These appliances together will consume 325.8W power. If you take a 400W solar inverter, it will give a power backup of up to 320W, considering all the core and copper losses. If you don't use a computer, the power consumption reduces to only 245.9 watts.

How many solar panels do you need for your home

To calculate solar power needs of your home, proceed as follows:

1. How much power do you use per day? To calculate this, get your last one year's monthly power bills and calculate your average kilowatt hour (kWh) usage per month. The reason why we have considered the 12-month time period is because our power consumption fluctuates with the seasons. (In case you don't have all your power bills, simply use the last month's.) Now take your average monthly usage and divide it by 30 to work out your average daily power consumption. As we get our electricity bill in units, we have to convert units into watts.

Let's say the electricity consumption of a house is 168 units per month. So daily power usage is $168/30=5.6$ units. As 1 kWh=1 unit, the power consumption in watts would be 5.6 kWh.

2. Calculate the total wattage of solar power needed. Before you can work out number of solar panels you may need, you need to find out how many usable hours of sunlight your region gets each day. A simple way to find that out is to have a look at an isolation map. You could just Google to find one.

Now take the average daily kWh calculation and divide that by the number of daily usable sunlight hours, then multiply that by 1.25 (to take into account the wasted energy from wiring, charge controllers, batteries and inverters).

Continuing from our example, our solar panel watt needs equal:

$5.6 \text{ kWh needed} / 8 \text{ hrs of sunlight}$

$\times 1.25 = 0.875 \text{ kW or } 875 \text{ watts a day}$

If you want to use a 100W solar panel, you will need $875/100=8.75$, i.e., 9 pieces of 100W monocrystalline solar panels.

This indicates that your home solar power system must have the capacity to produce a minimum of 875-watt power. (The backup of system is based on battery capacity.)

ZMR—backup analysis for batteries

The battery backup for the required load can be calculated by Z-factor, where $Z=0.2222$. This Z-factor has been found out from the observations with batteries used for certain loads. As the wattage increases, this 'Z' value also increases. Z-factor decreases with aging of the battery.

TABLE III
Supply to LCD Computer with 12V, 120Ah Battery, Solar Inverter without UPS

Time	Output voltage (Vo)	Output current (Io)	Battery voltage
Initially	230V	10A	12.3V
Initially	230V	11A-13A	12.3V
Booting time	230V	12A-14A	12.3V
Running time	230V	13A	12.2V
11 minutes	230V	12.5A	12.1V-12.3V
17 minutes	230V	13.5A	12.2V-12.3V
1:02 hours	230V	13A	12.2V-12.3V
1:35 hours	220V	12A-14A	12.1V-12.3V
2:02 hours	220V	13A	12.1V
2:32 hours	220V	13A	12.0V-12.1V
2:42 hours	220V	12.5A	12.0V
3:43 hours	220V	13A-15A	11.9V
3:43 hours	220V	13A-15A	11.1V-11.2V
4:00 hours	220V	13A-15A	11.0V-11.1V